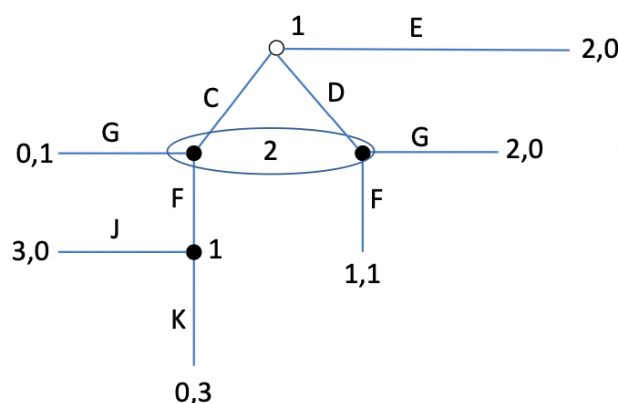


TØ 7 – Mikroøkonomi 2 – 2024

Problem 1

Problem 1

Consider the following extensive form game:



- Does the game have a perfect Bayesian equilibrium C and/or D with positive probability? Explain.
- Find the perfect Bayesian equilibria of the game.

a)

Define α to be the probability player 1 puts on C and β the probability he puts on D .

This implies that G is optimal for player 2 if $\alpha > \beta$ and if $\alpha < \beta$, F is optimal for player 2 since if $\alpha > \beta$, the probability of reaching C is highest, making player 2 choose G , as this will get him a payoff of 1. Had player 2 chosen F in this scenario, player 1 would have chosen J , leaving player 2 with a payoff of 0. If $\alpha = \beta$, a mix between F and G will occur. It is not a PBE: for $\alpha > \beta$, player 2 chooses G , player 1 will deviate by choosing D , since here, player 1 can get a payoff of 2 rather than 1 from the C, G path, so there is not an equilibrium with positive probability on D .

If player 2 instead chooses F (i.e., in the $\alpha < \beta$ scenario), player 1 will choose C , and J afterwards, which would not be optimal for player 2, so there is not an equilibrium with a positive probability of C either.

Finally, if player 2 mixes between F and G ($\alpha = \beta$ scenario), player 1 will put a positive probability on D. However, as player 1 can get a payoff of either 1 or 2 here, but 2 for sure by choosing E, it is not an equilibrium to mix.

To sum up, there is no PBE with positive probabilities on C nor D

b)

First, notice that player 2 will in equilibrium always play E and J, as explained in part a. Player 2 will prefer G when the expected payoff of playing G is greater than F:

$$\mathbb{E}(u_2(G)) = 1 \cdot \mu + 0(1 - \mu) \geq 0 \cdot \mu + 1(1 - \mu) = \mathbb{E}(u_2(F)) \Leftrightarrow \mu \geq \frac{1}{2}$$

Which is the first PBE

Second PBE: here, player 2 is indifferent between F and G, i.e., $\mu = \frac{1}{2}$

Probability ρ of playing F (since if $\rho > \frac{2}{3}$, player 1 will have an incentive to deviate by playing C, which would destroy the equilibrium).

$$1 \cdot \rho \leq 2 \cdot (1 - \rho) \Leftrightarrow \rho \leq \frac{2}{3}$$

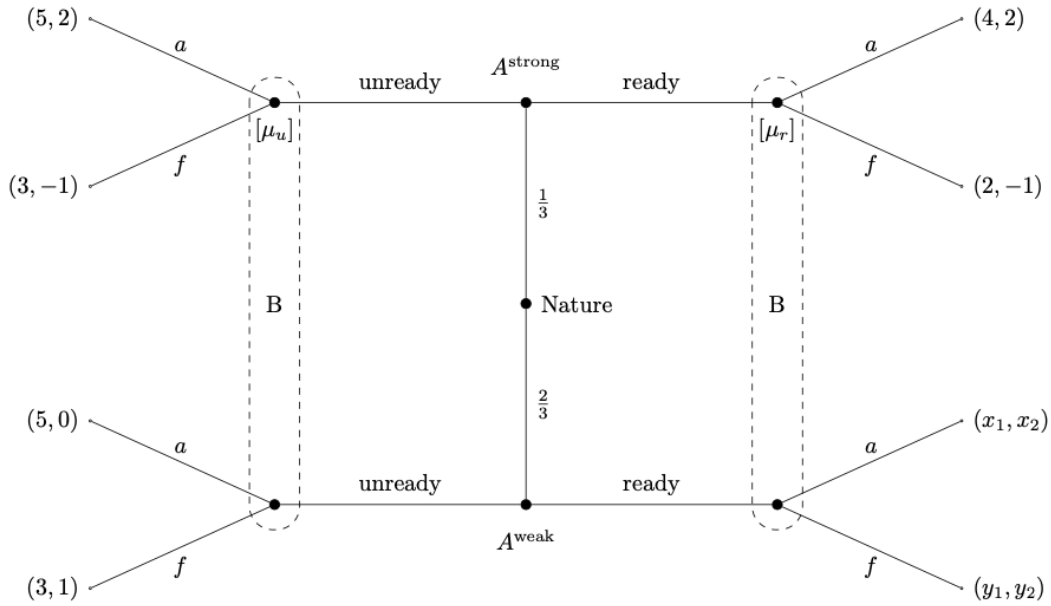
The PBEs are then:

$$PBE = \left\{ \left(EJ; G, \mu \geq \frac{1}{2} \right); \left(EJ; \rho, 1 - \rho, \mu = \frac{1}{2} \text{ and } \rho \leq \frac{2}{3} \right) \right\}$$

Why is there not a PBE where player 2 prefers F over G when we had one where he preferred G over F? Because if he preferred F over G, player 1 would always choose C and then J, making it not an equilibrium

Problem 2

Newcomer firm A plans to enter a market dominated by incumbent firm B . With probability $\frac{1}{3}$, A has received venture capital making it *strong*. With probability $\frac{2}{3}$, it has not succeeded in attracting capital and is therefore *weak*. A knows its type, but B does not. Firm A can either *ready* itself for battling or remain *unready*. Firm B observes whether A is ready or not and chooses whether to fight (f) or to acquiesce (a). The following game tree describes the situation.

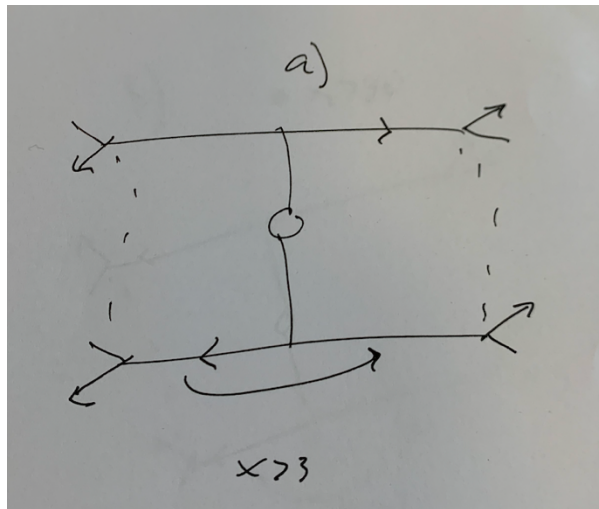


- For what values of x_1, x_2, y_1 , and y_2 , if any, does the game have a separating equilibrium in which a strong newcomer chooses *ready* and a weak newcomer chooses *unready*?
- For what values, if any, does the game have a pooling equilibrium in which both types of newcomer choose *unready*?

Note: player 1 is player A and player 2 is player B

a)

A separating equilibrium if $x_1 \leq 3$ since if not, player 1 weak could deviate by playing ready, getting a payoff larger than before



PBE is

$$((ready, unready); (f, a), \quad x_1 \leq 3)$$

b)

Expected utility of player 2 given Nature's probabilities $\mu_u = \frac{1}{3}, 1 - \mu_u = \frac{2}{3}$

$$\mathbb{E}U_2[a|\mu_u] = 2\mu_u + 0(1 - \mu_u) = \frac{2}{3}$$

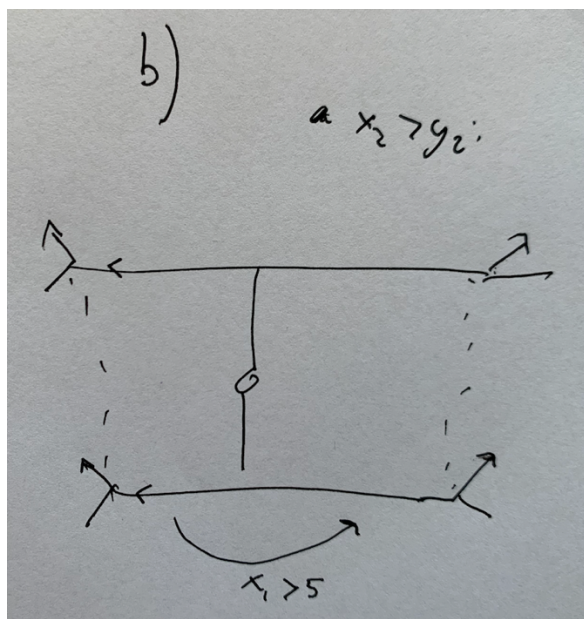
$$\mathbb{E}U_2[f|\mu_u] = -1\mu_u + 1(1 - \mu_u) = \frac{1}{3}$$

So, $\mathbb{E}U_2[a|\mu_u] > \mathbb{E}U_2[f|\mu_u]$ and therefore, player 2 should choose a

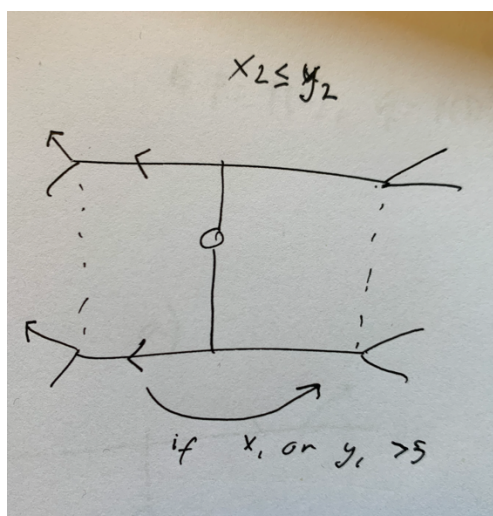
Now, considering when A is strong, i.e., in the RHS of the figure

If $x_2 > y_2$, player 2 will always choose a since this strictly dominates f as $2 > -1$ (upper RHS of the figure)

This would be an equilibrium, if and only if $x_1 \leq 5$ since if $x_1 > 5$, a weak player 1 could deviate to ready and get a payoff greater than her equilibrium payoff



$x_2 \leq y_2$: If this is the case, player 2 will mix with some values μ_r and $1 - \mu_r$ (we cannot calculate these values as we do not have any payoffs in weak, ready scenario). Since player 2 mixes, A weak could deviate if either the payoff x_1 or y_1 was greater than the payoff player 1 got in weak, unready (left bottom part of figure).



So, in conclusion the PBE are (player A's strategies are read top to bottom and player 2's left to right)

From part a:

$$((ready, unready); (f, a), \quad x_1 \leq 3)$$

And from part b:

$$((unready, unready); (a, a), x_2 > y_2, x_1 \leq 5)$$

The other PBE from b:

$$((unready, unready); (a, x), \min x_1, y_1 \leq 5)$$

Problem 3

Consider the following game between a worker and two employers. The order of play is as follows:

- Nature chooses the worker's ability $a \in \{2, 5.5\}$, the *Low* and *High* ability each having probability 0.5. The variable a is observed by the worker, but not by the employers.
- The worker chooses education level $s \in \{0, 1\}$.
- The employers each offer a wage contract $w(s)$.
- The worker accepts a contract, or rejects both of them.
- Output equals a .

The worker's payoff is his wage minus his cost of education, and the employer's is his profit.

$$\pi_{worker} = \begin{cases} w - 8s/a & \text{if the worker accepts contract } w. \\ 0 & \text{if the worker rejects both contracts.} \end{cases}$$

$$\pi_{employer} = \begin{cases} a - w & \text{for the employer whose contract is accepted.} \\ 0 & \text{for the other employer.} \end{cases}$$

- (a) Find the pooling equilibrium of this game. What are the payoffs of the worker's types (and the payoffs averaged across the worker's types) in each equilibrium?
- (b) Find the separating equilibrium of the game. [Hint: Competition between the employers makes the participation constraints hold as equalities.]
- (c) Does the pooling equilibrium satisfy the Intuitive Criterion? Explain why or why not.

a)

In pooling, the employer does not know whether a worker is high or low ability and therefore, it is optimal to choose education level: $s_L = s_H = 0$ and the employer pays the wage $\frac{2+5.5}{2} = 3.75$. This is an equilibrium since no party would bother to choose education, since the type is unobservable for the employer

Also, there does not exist a pooling equilibrium on $s_L = s_H = 1$ since the low type would deviate

b)

Separating equilibrium

$$s_L = 0, s_H = 1, w(0) = 2, w(1) = 5.5$$

Must satisfy the ICL and ICH

$$U(s_L = 0|L) \geq U(s_L = 1|L)$$

Which is satisfied for the low type as $a = 2$:

$$w(0) - \frac{8 \cdot 0}{a} \geq w(1) - \frac{8 \cdot 1}{a} \Rightarrow 2 \geq 5.5 - 4 = 1.5$$

And ICH

$$U(s_H = 1|H) = 1 \geq U(s_H = 0|H) = 0$$

Satisfied since ($a = 5.5$)

$$w(1) - \frac{8 \cdot 1}{a} \geq w(0) - \frac{8 \cdot 0}{a} \Rightarrow 4.05 \geq 2$$

c)

The pooling equilibrium violates the intuitive criterion since in this equilibrium, if a person is educated, the employer believes that he/she is a high type. This would make the high ability workers choose education, which then translates into the equilibrium discussed in part b

Problem 4

Consider again the card game that we studied in the lecture, with the following variation: Each player still puts \$1 into the pot. But now *each* player is dealt a card; each player's card is equally likely to be *High* or *Low*, independent of the other player's card. Each player only sees his own card. Player 1 may *see* or *raise*.

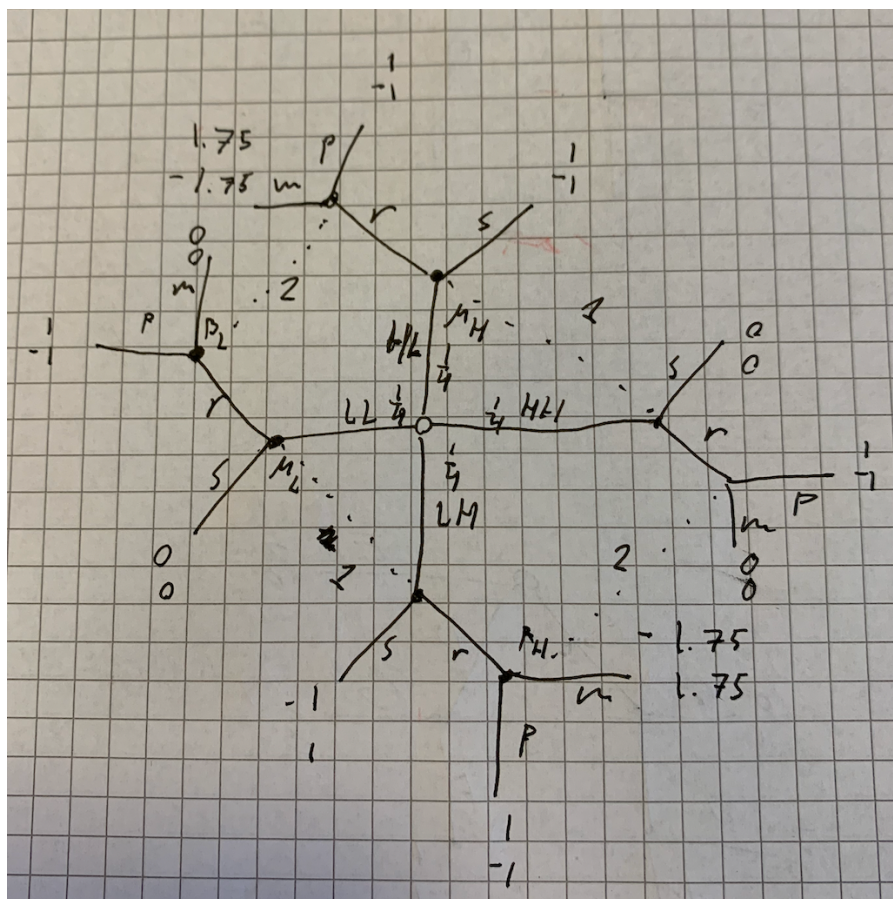
If he *sees*, then the players compare their cards. The one with the higher card wins the pot; if the cards are the same, then each player takes back the dollar he put in the pot.

2

If player 1 *raises*, then she puts an additional \$0.75 to the pot, and player 2 may *pass* or *meet*. If player 2 *passes*, then player 1 takes the money in the pot. If player 2 *meets*, then she also puts an additional \$0.75 into the pot, and players compare their cards. The one with the higher card wins the pot; if the cards are the same, then each player takes back the $\$(1 + 0.75)$ he/she had put in the pot.

- (a) Model this card game as an extensive game. (Hint: By putting the initial node into the very center of your diagram you avoid that information sets would cross.)
- (b) What are the strategies of the players in this game?
- (c) Find the Perfect Bayesian equilibrium or equilibria of the game.

Part a



Part b

Player 1 has the strategy sets

$$\{ss, sr, rs, rr\}$$

And player 2

$$\{mm, mp, pm, pp\}$$

Part c

Payoffs are calculated where first letter of player 1 is her strategy if her card is High and second letter is her strategy if card Low. The same principle for player 2

For example:

(sr, mm)

When player 1's card High: 1 and 0

And when Low: 0 and -1.75

Now times the probabilities, $\frac{1}{4}$:

$$\frac{1}{4}(1 + 0 + 0 - 1.75) = -0.1875 = -\frac{3}{16}$$

Another example:

(rs, pm) payoff of player 1

When high: 1.75 and 1

Low: 0 and -1

$$\frac{1}{4}(1.75 + 1 + 0 - 1) = 0,4375 = \frac{7}{16}$$

$1 \backslash 2$	mm	mp	pm	pp
ss	0, 0	0, 0	0, 0	0, 0
sr	$-\frac{3}{16}, -\frac{3}{16}$	$\frac{1}{16}, -\frac{1}{16}$	$\frac{1}{2}, \frac{1}{2}$	$\frac{3}{4}, -\frac{3}{4}$
rs	$\frac{3}{16}, -\frac{3}{16}$	0, 0	$\frac{7}{16}, -\frac{7}{16}$	$\frac{1}{4}, -\frac{1}{4}$
rr	0, 0	$\frac{1}{16}, -\frac{1}{16}$	$\frac{15}{16}, -\frac{15}{16}$	1, -1

Now, use IESDS:

$$sr \geq rr$$

$$mm \geq pm$$

$$mm \geq pp$$

$$rs \geq ss$$

Now, the MSNE

		q	$(1-q)$
	$1/2$	rr	rp
P	rs	$\frac{3}{16}, \frac{3}{16}$	$0, 0$
$(1-P)$	rr	$0, 0$	$\frac{1}{16}, \frac{1}{16}$

Player 1 indifferent when

$$\frac{3}{16} \cdot q + 0 \cdot (1-q) = 0 \cdot q + \frac{1}{16} \cdot (1-q) \Leftrightarrow q^* = \frac{1}{4}$$

Player 2:

$$-\frac{3}{16} \cdot p + 0 \cdot (1-p) = 0 \cdot p + \frac{1}{16} \cdot (1-p) \Leftrightarrow p^* = \frac{1}{4}$$

Bayes Rule now

$$\beta_H = \frac{p \cdot (1-p)}{p \cdot (1-p) + p \cdot 1} = \frac{\frac{1}{4} \left(1 - \frac{1}{4}\right)}{\frac{1}{4} \left(1 - \frac{1}{4}\right) + \frac{1}{4} \cdot 1} = \frac{3}{7}$$

$$\beta_L = \frac{q \cdot (1-q)}{q \cdot (1-q) + q \cdot 1} = \frac{\frac{1}{4} \left(1 - \frac{1}{4}\right)}{\frac{1}{4} \left(1 - \frac{1}{4}\right) + \frac{1}{4} \cdot 1} = \frac{3}{7}$$

So,

$$\beta_H = \beta_L$$

And the μ 's

$$\mu_H = \mu_L = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \frac{1}{2}$$